CAPSTONE SHOWCASE

Moringa School
Data Science Track
CRISP-DM Methodology
LightGBM Classifier

Predicting Contraceptive Use Among Kenyan Women

A Machine Learning Approach to Targeting Family Planning Interventions Using the Kenya Demographic and Health Survey 2022

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Project Overview & Target Distribution

The Core Problem

Kenya's national modern contraceptive prevalence rate (mCPR) is approximately **60%**, but this hides severe regional and socioeconomic disparities. Our goal is to predict modern contraceptive use to help health planners target resources effectively.

Target Variable Distribution

Most women fall into just two groups: **"No method" (18,694)** and **"Modern" (12,195)**. Traditional (1,214) and Folkloric (53) methods are much rarer. For our model, we target predicting **Modern vs. Not-modern**.

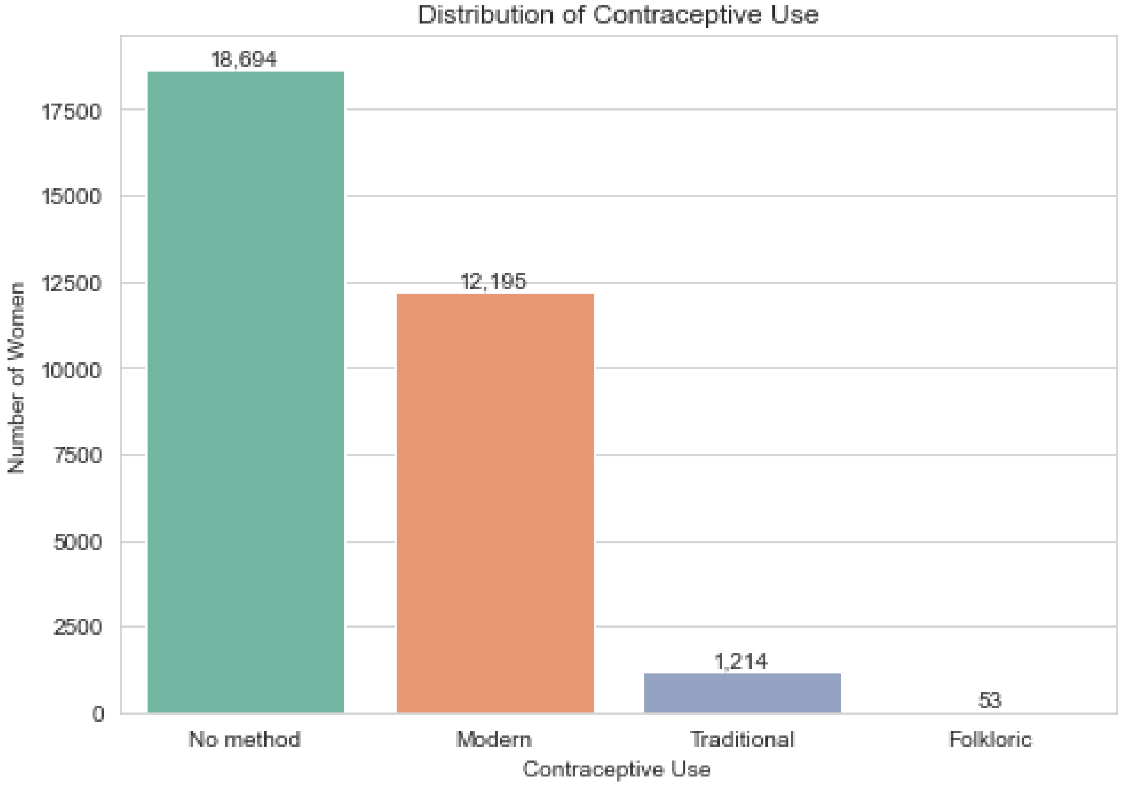

Figure: Distribution of Contraceptive Use

Demographics

Age Distribution

Most of the women are in their twenties (average age about 29, median 28), and the numbers thin out towards 49. This represents the primary reproductive age demographic accurately.

Figure: Age Distribution

Education Level

Primary and secondary are the most common levels (~37% each). Higher education is about 15%, while roughly 12% of the women have no formal education.

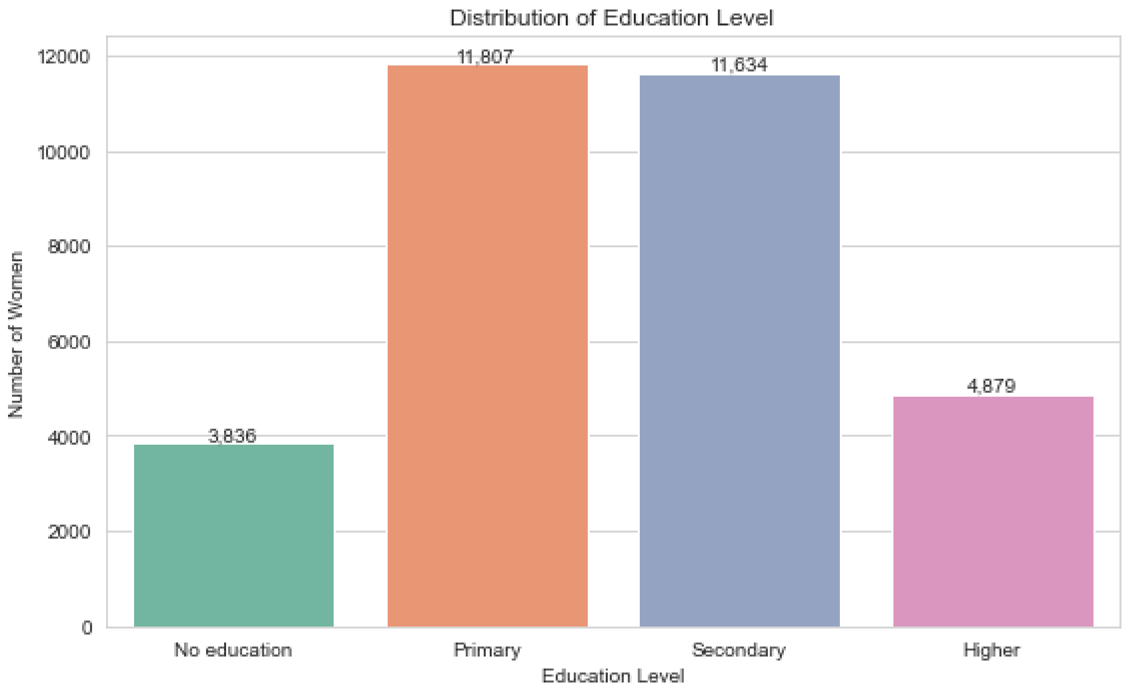

Figure: Education Level

Socioeconomics

Wealth Index

The five wealth groups are fairly evenly represented (each is roughly 18 to 22% of the sample), providing sufficient data across all economic statuses.

Figure: Wealth Index

Residence Type

About 62% of the women live in rural areas and 38% in urban areas, reflecting Kenya's overall demographic split.

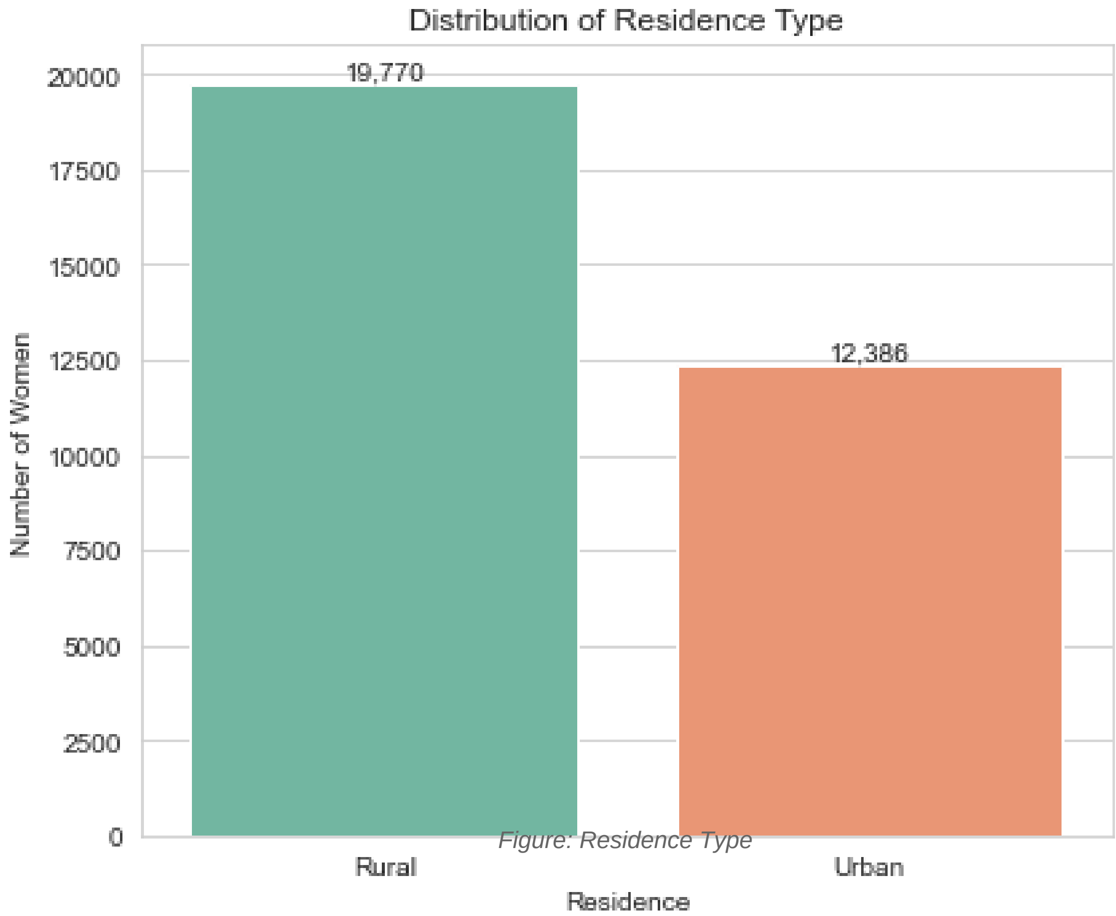

Figure: Residence Type

Bivariate Analysis

Education as a Driver

The clearest gap in modern contraceptive use is between women with no education (only ~12% use a modern method) and women with any education (~37% to 46%). Education acts as a major catalyst for uptake.

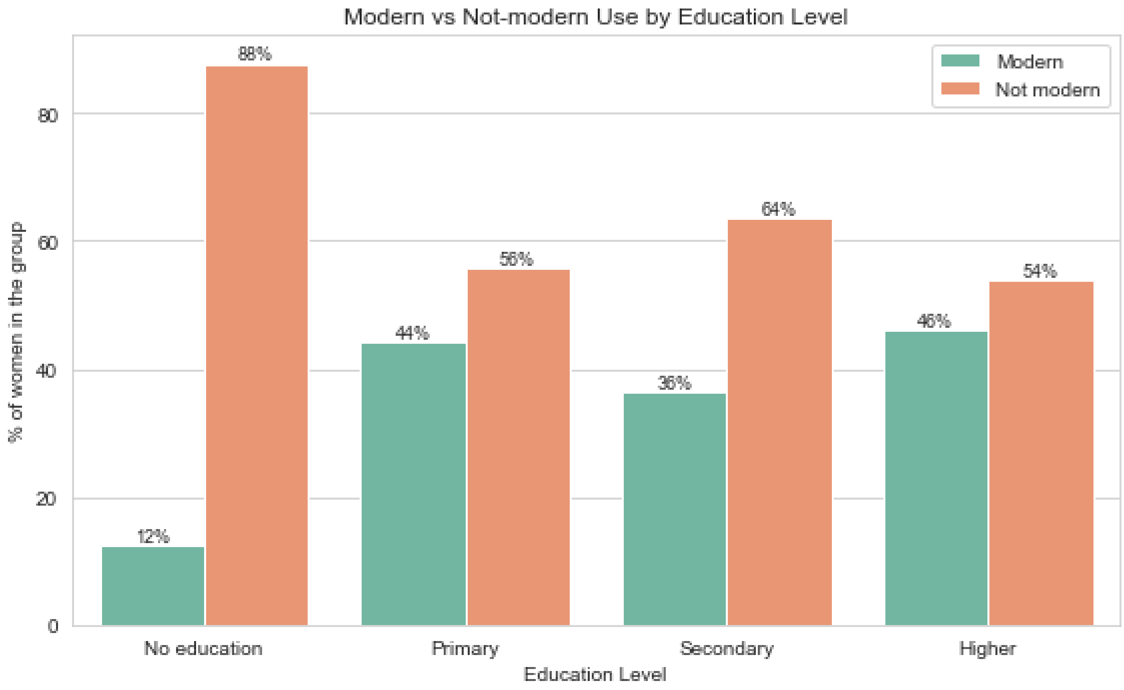

Figure: Use by Education

Wealth as a Driver

The poorest women stand out with the lowest modern use (~25%). Above the poorest group, the rate is fairly flat at around 41% to 43%, showing that while extreme poverty is a barrier, wealth alone doesn't scale usage endlessly.

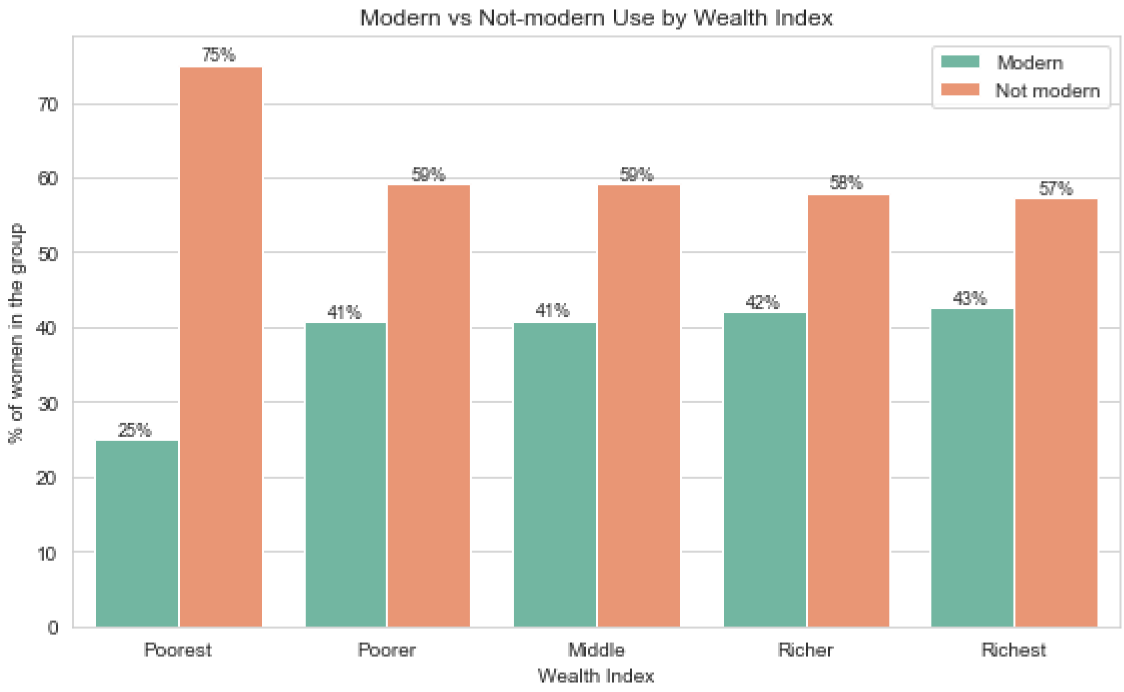

Figure: Use by Wealth

Geographic Disparities

Modern Use by County

County makes a massive difference. Modern use runs from about **1% in Mandera** and under 6% in Wajir, Garissa, and Marsabit (all in the arid north) up to **over 50% in central/eastern counties** like Embu, Meru, Nyeri, and Kirinyaga.

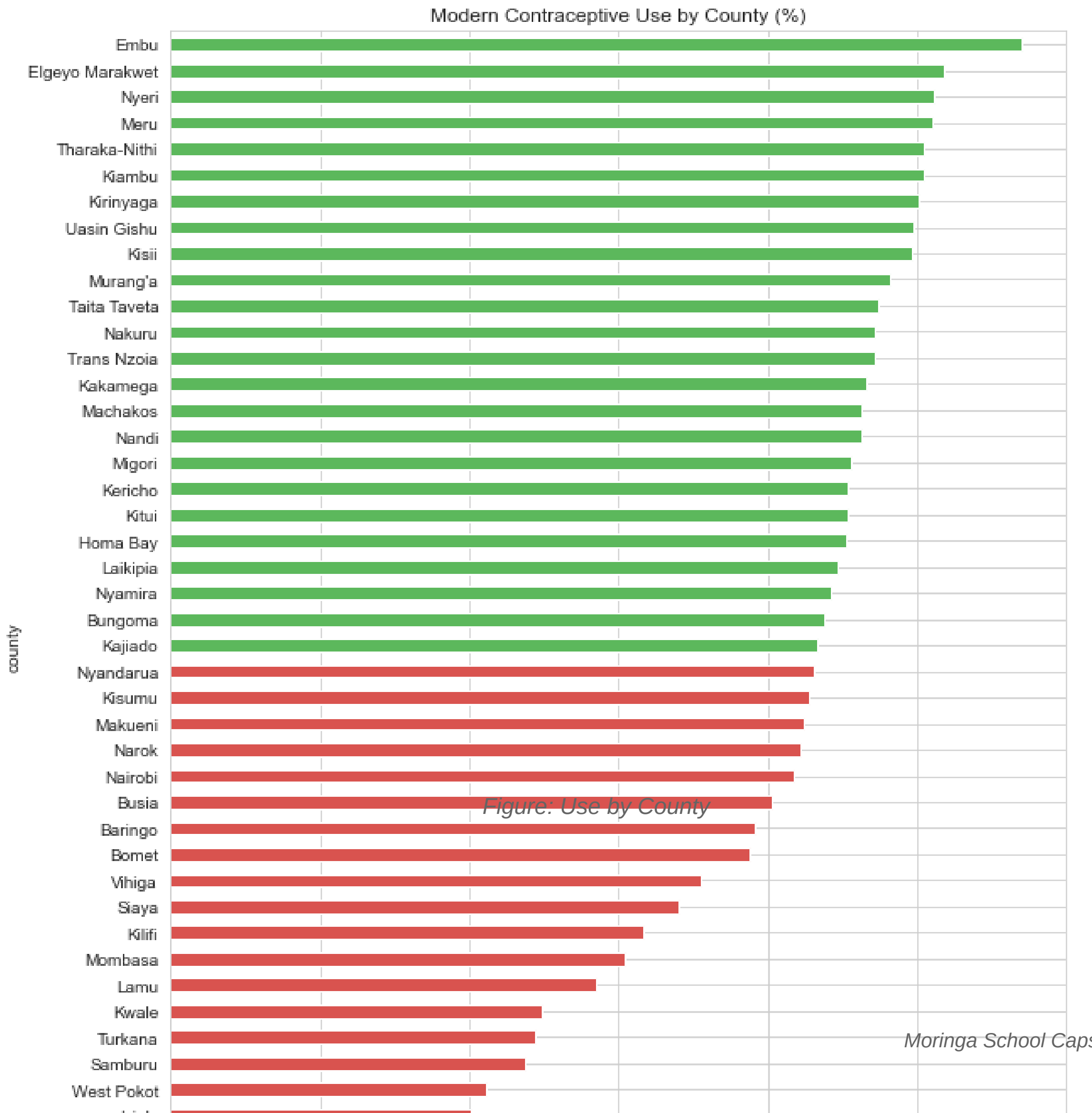

Figure: Use by County

Multivariate Analysis

Does Wealth overcome a lack of Education?

Reading across the heatmap, **education matters significantly more than wealth**. Among women with no education, modern use stays low (about 9% to 21%) no matter how wealthy their household is. Conversely, highly educated women in the poorest households still adopt modern methods at a much higher rate (~36%).

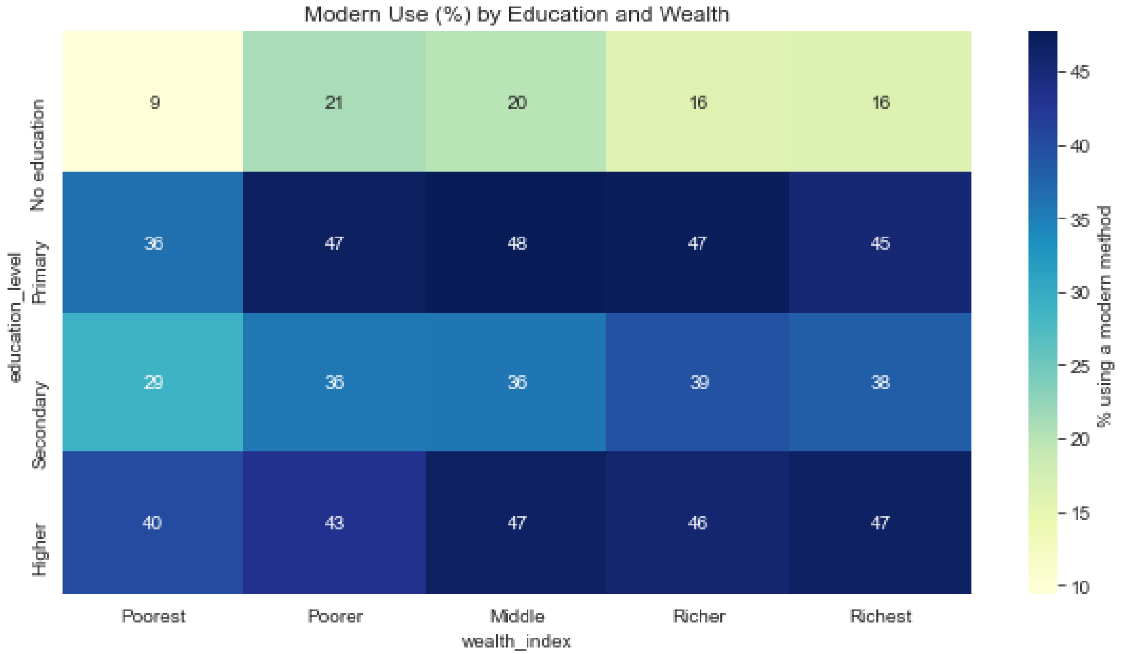

Figure: Education and Wealth Interaction

Data Preprocessing & Modeling Strategy

Preprocessing & Leakage Control

- **Structural Missingness:** Imputed `0` for `age_first_birth` for women with no children, creating a `has_given_birth` flag.
- **Leakage Control:** Excluded behavioral target variables like `ever_used_contraceptive`.
- **SMOTE:** Applied Synthetic Minority Over-sampling Technique exclusively inside cross-validation folds to prevent data leakage and handle class imbalance.

Iterative Modeling (Cross-Validation)

Model	Mean CV ROC-AUC	Mean CV Accuracy
Logistic Regression (Baseline)	0.7962	71.6%
Random Forest	0.8150	72.6%
XGBoost	0.8141	73.4%
LightGBM (Tuned)	**0.8189**	**73.6%**

Final Model Evaluation

We selected **LightGBM (Tuned)** as our final model for its superior discriminative power and high recall for the minority class.

Hold-out Test Performance

- **Test ROC-AUC:** `0.8260`
- **Test Accuracy:** `74.4%`
- **Test Recall (Class 1):** `77.2%` (Crucial for public health targeting, minimizing false negatives)

ROC Curve Comparison

The ensemble tree-based models significantly outperformed the Logistic Regression baseline, proving that non-linear relationships and interactions are key in this dataset.

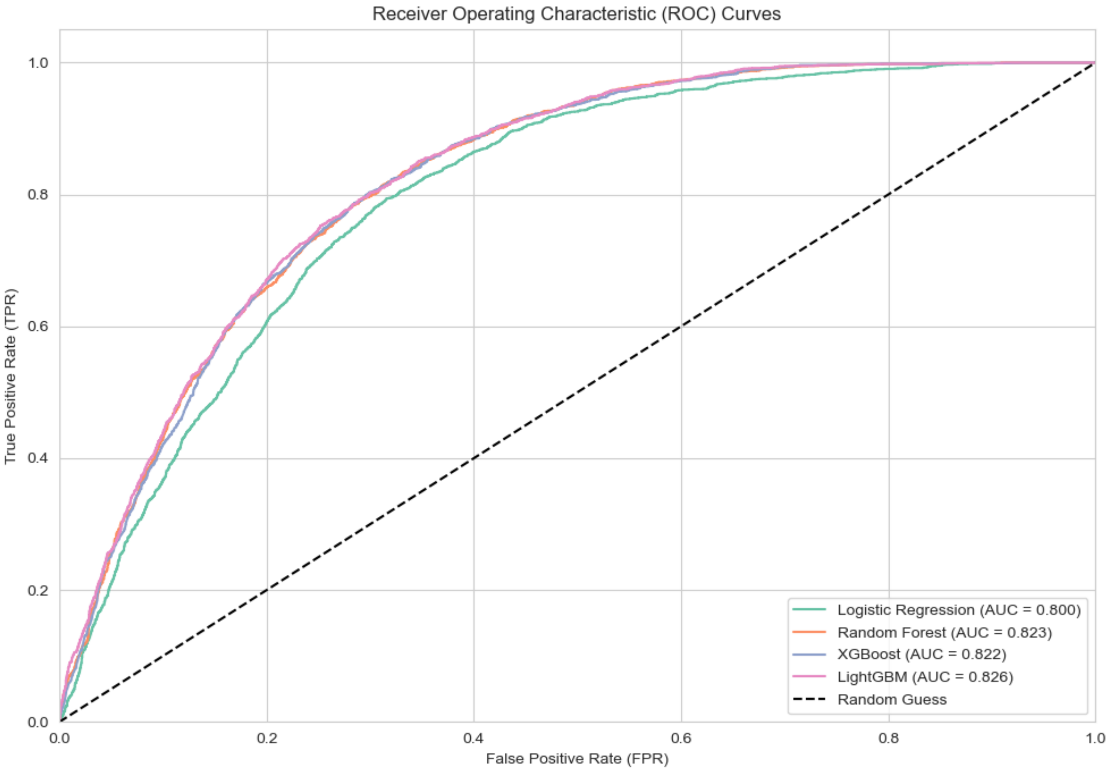

Figure: ROC Curves

Model Evaluation Details

Confusion Matrix Insights

Our model correctly identified the vast majority of true modern contraceptive users. High recall is preferred here because missing an opportunity to intervene (False Negative) is costlier in public health than reaching out to someone who is already using a method (False Positive).

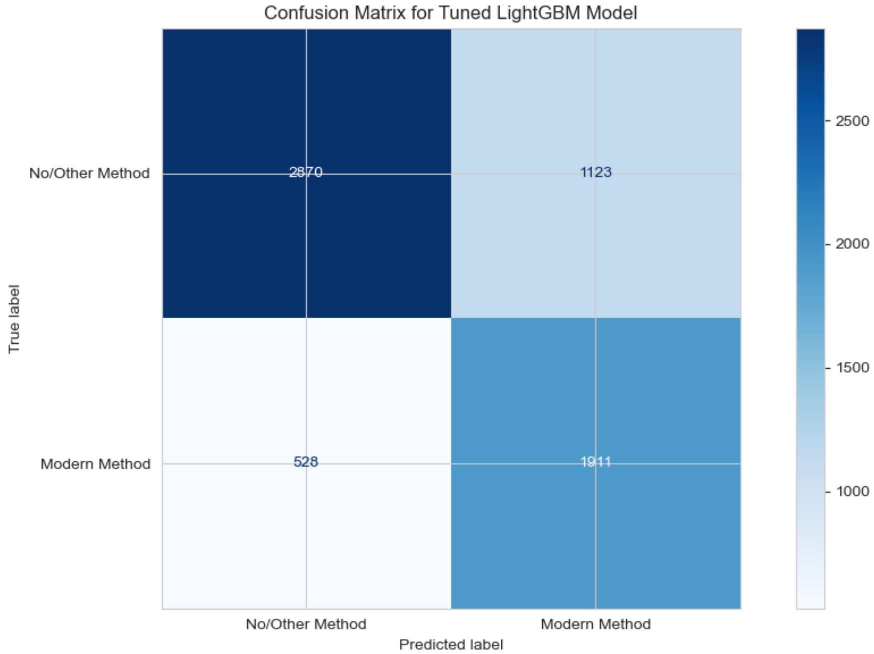

Figure: Confusion Matrix

Interpretability

Using SHAP (SHapley Additive exPlanations), we broke down exactly what drives the model's predictions.

Key Drivers

1. **Education Level:** The strongest individual-level predictor.
2. **County Constraints:** Living in ASAL regions acts as a massive penalty, heavily dragging down adoption probabilities regardless of individual traits.
3. **Parity (Living Children):** Uptake jumps significantly after a woman has 1-2 children, signaling a shift from spacing to limiting births.

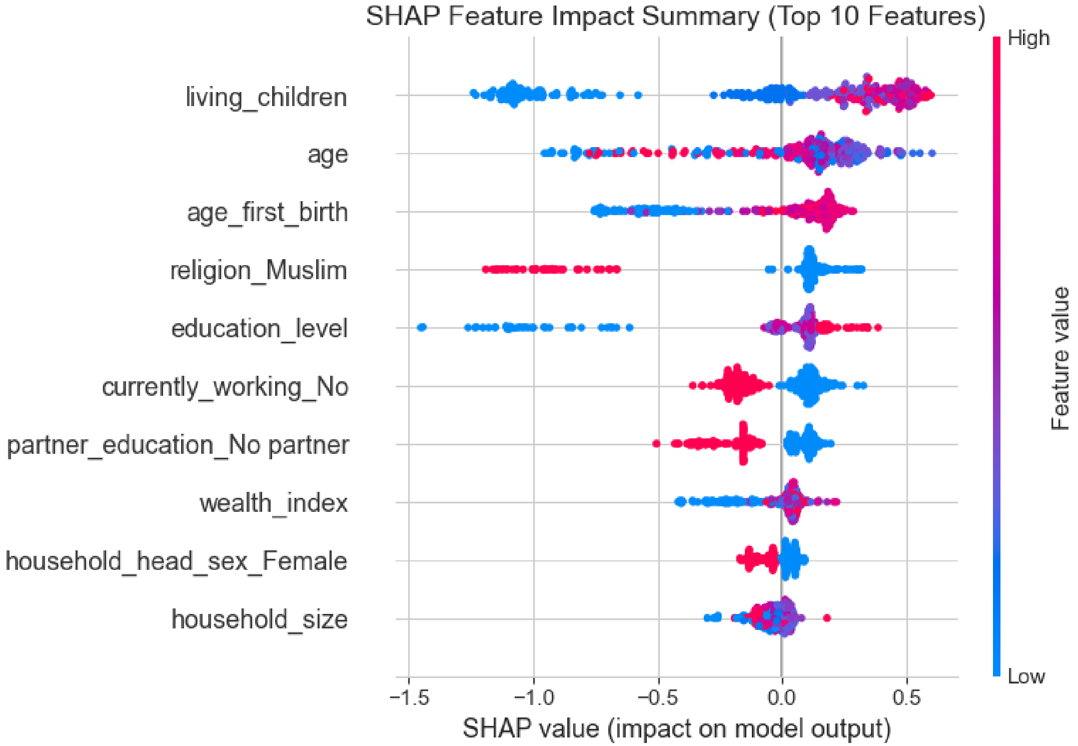

Figure: SHAP Summary Plot

Web Application Deployment

To make our model accessible to stakeholders (Ministry of Health, CHPs), we deployed it as an interactive web application.

Architecture

- **Backend:** Flask API loading our serialized `contraceptive_pipeline.joblib` (which encapsulates both the preprocessing `ColumnTransformer` and the `LightGBM` model).
- **Frontend:** A beautiful, responsive glassmorphic UI built with HTML/CSS.
- **Functionality:** Users can input demographic data and receive an instant probability score for modern contraceptive adoption, along with dynamic policy recommendations.

Actionable Business & Policy Recommendations

Based on our EDA and SHAP explainability insights, we propose three core interventions:

1. Geographic Prioritization in ASAL Regions

Counties like Mandera, Wajir, and Garissa face systemic barriers (cultural, infrastructural) pulling adoption down to <6%. Direct funding to mobile clinics and reliable supply chains in these specific counties is essential.

2. Outreach Targeting Low-Education Clusters

Education is the strongest individual predictor. Community Health Promoters (CHPs) should prioritize direct household visits and counseling in areas where average female literacy is low, substituting formal education with targeted health literacy campaigns.

3. Subsidize the "Poorest" Quintile

The poorest wealth quintile sees a steep drop-off in usage (~25%). Interventions must ensure contraceptive methods are entirely free and accessible to bridge the poverty gap.

Conclusion & Q&A

Thank You!

Questions?

Predicting Contraceptive Use in Kenya

A Moringa School Capstone Project by The Insight Architects

- **GitHub Repository:** [Project Repo Link](https://github.com/symekah1999/Predicting-contraceptive-use-among-women_Capstone-Project-The-Insight-Architects-)
- **Serialized Model:** `contraceptive_pipeline.joblib`
- **Flask App:** `app.py` (Local Port `5000`)